The Macroeconomics of Large Language Model Scaling: Inference Costs, Capital Obsolescence, and Bubble Dynamics

Dr. G. R. Rajkumar
RV College of Engineering
Bengaluru, India
rajkumargr@rvce.edu.in

Sankalp Khamesra
RV College of Engineering
Bengaluru, India
sankalpk.ai23@rvce.edu.in

L Moryakantha
RV College of Engineering
Bengaluru, India
lmoryakantha.ai24@rvce.edu.in

Sachit Ramesha Gowda
RV College of Engineering
Bengaluru, India
sachitramesha.ai23@rvce.edu.in

Preetham R
College of Engineering
Bengaluru, India
preethamr.ai23@rvce.edu.in

***Abstract*—The rapid scaling of large language models (LLMs) has triggered an unprecedented reallocation of capital toward artificial intelligence infrastructure, semiconductor manufacturing, and cloud computing platforms. Equity markets have responded with extraordinary valuation growth for firms positioned at critical nodes of the AI supply chain, often outpacing observable adoption rates and measured productivity gains in the real economy. This paper examines the macroeconomic implications of LLM scaling through the lens of capital accumulation, valuation dynamics, and speculative behavior. We synthesize empirical evidence from historical valuation metrics, market concentration data, and enterprise adoption trends with a formal analytical framework linking compute-driven scaling laws to expected economic surplus. By deriving a necessary condition under which AI-related asset prices decouple from realizable productivity growth, we identify structural features that render the AI economy susceptible to bubble formation. Comparative analysis with prior technology-driven booms highlights both similarities—such as narrative-driven expectations and capital concentration—and novel risks arising from extreme compute intensity and supply-side constraints. The results contribute to a deeper understanding of how frontier AI technologies propagate through financial markets and inform policy discussions on systemic risk, innovation financing, and long-run productivity growth.**

Keywords—Large language models, artificial intelligence economics, asset bubbles, capital concentration, compute scaling, macroeconomic modeling, technology adoption, financial valuation

I. Introduction

The rapid commercialization of large language models (LLMs) has initiated a structural transformation in global capital allocation. Unlike earlier waves of digital innovation, LLMs are fundamentally constrained by computational scale, specialized semiconductor hardware, energy-intensive data centers, and vertically integrated cloud infrastructure. Since 2022, these constraints have driven unprecedented levels of capital expenditure and valuation growth within a narrow segment of the technology sector, raising renewed concerns regarding speculative dynamics and the long-run sustainability of asset prices [1]–[4].

Equity market evidence indicates that firms positioned at critical nodes of the artificial intelligence (AI) supply chain—semiconductor manufacturers, hyperscale cloud providers, and frontier model developers—now account for a historically large share of aggregate market capitalization and index-level returns [5]–[7]. Analysis of S&P 500 concentration metrics shows that the so-called Magnificent Seven firms contribute a level of index dominance comparable to, and in some measures exceeding, that observed during the late-1990s dot-com boom [8], [9].

Figure 1 illustrates this concentration by showing the growing share of total S&P 500 market capitalization held by the Magnificent Seven relative to the remainder of the index from 2015 onward. The figure highlights the extent to which recent equity market performance has become increasingly dependent on a small subset of AI-exposed firms, amplifying systemic exposure to valuation shocks within this group.

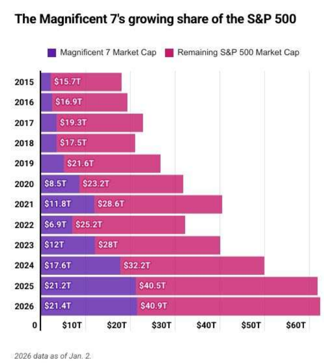

Figure 1: Growing share of total S&P 500 market capitalization accounted for by the "Magnificent Seven" firms, illustrating increasing equity market concentration. Source: Motley Fool; S&P Dow Jones Indices.

Beyond concentration effects, long-horizon valuation indicators suggest a widening divergence between asset prices and underlying economic fundamentals. Cyclically adjusted price-to-earnings (CAPE) ratios, Tobin's q, and market-capitalization-to-GDP metrics place current U.S. equity valuations in ranges historically associated with elevated correction risk [10]–[13]. Figure 2 presents the evolution of the Shiller CAPE ratio, calculated as the average of inflation-adjusted earnings over the previous ten years. Historically, CAPE values exceeding 30 have served as a harbinger of speculative excess, followed by severe market drawdowns. As shown in the figure, the CAPE ratio rose above 30 in late 2023 and remained elevated thereafter, surpassing 40 by late 2025—an occurrence that has happened only a handful of times since 1871. Previous episodes at comparable valuation levels include the 1929 stock market peak preceding an 89% decline in the Dow Jones Industrial Average and the dot-com bubble, after which the S&P 500 lost nearly half of its peak value.

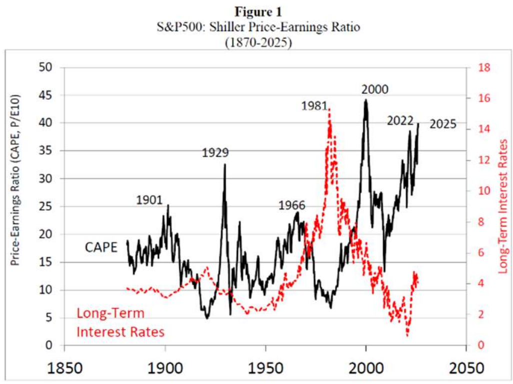

Figure 2: Historical evolution of the S&P 500 Shiller cyclically adjusted price–earnings (CAPE) ratio, highlighting periods of elevated valuation and speculative excess. Source: ShillerData

While elevated valuations may reflect expectations of transformative productivity gains from AI, economic history demonstrates that general-purpose technologies (GPTs) typically generate delayed and uneven productivity effects. Such gains require complementary organizational restructuring, institutional adaptation, and demand expansion before diffusing broadly across the economy [14]–[17]. As a result, rapid valuation growth in advance of realized productivity improvements has frequently coincided with speculative dynamics rather than sustainable growth.

Evidence from enterprise adoption studies further complicates the prevailing valuation narrative. Although experimental use of LLM-based tools is widespread across firms of all sizes, multiple surveys and industry analyses indicate that large organizations are experiencing slowing adoption, rising integration costs, governance challenges, and diminishing short-run returns on AI investment [18]–[21]. Figure 3 illustrates enterprise AI adoption rates disaggregated by firm size, showing that adoption among large firms has begun to plateau or decline even as smaller firms continue to increase usage. This divergence suggests frictions in scaling AI technologies within complex organizational environments, calling into question assumptions of rapid, economy-wide productivity diffusion.

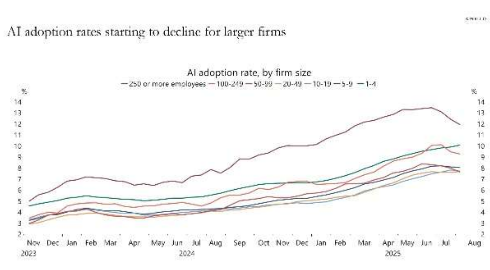

Figure 3: Enterprise AI adoption rates over time by firm size, showing a recent plateau and decline in adoption among large firms. Source: Apollo Academy

`A defining characteristic of the current AI cycle is its extreme capital intensity. Frontier LLM development requires sustained investment in advanced GPUs, custom accelerators, energy infrastructure, and long-term cloud capacity contracts, often secured through strategic cross-investments between AI developers, cloud service providers, and semiconductor manufacturers [23]–[26]. Figure 4 provides a schematic representation of these capital and compute interdependencies, mapping investment flows, service relationships, and hardware dependencies across the AI ecosystem. The dense network structure illustrated in the figure highlights how valuation shocks or demand shortfalls in one segment—such as cloud services or GPU supply—may propagate rapidly across the broader technology and financial system, increasing systemic fragility.

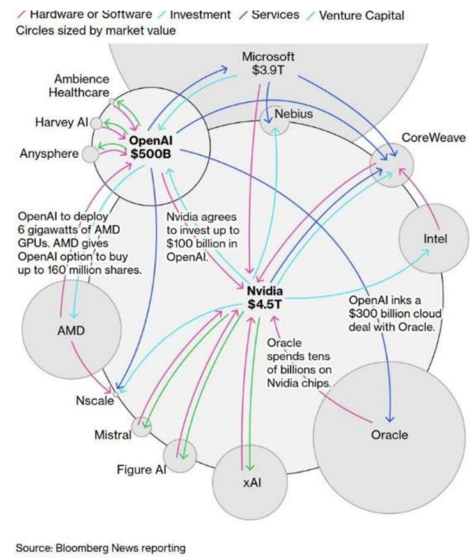

Figure 4: Capital, investment, and compute-service interdependencies among AI developers, cloud providers, and semiconductor manufacturers in the AI ecosystem. Source: Bloomberg News reporting.

Despite growing public discourse around an "AI bubble," most academic research has examined artificial intelligence through microeconomic or labour-market lenses, focusing on task automation, wage effects, and firm-level productivity [28]–[30]. In parallel, machine-learning research has established empirical scaling laws linking model performance to compute and data [31], [32]. However, these technical scaling relationships do not directly translate into macroeconomic surplus or sustainable profit growth, particularly in the presence of demand-side constraints and uneven adoption.

Recent macro-financial work, including the Institute for New Economic Thinking (INET) working paper by Storm [33], argues that contemporary technology-led growth narratives rely heavily on extrapolative expectations while underestimating distributional constraints, diminishing returns, and limits to aggregate demand. These mechanisms align closely with classical theories of speculative bubbles, which emphasize expectation feedback, narrative reinforcement, and capital concentration as drivers of asset price inflation disconnected from fundamentals [34], [35].

What remains absent from the literature is a unified macroeconomic framework that explicitly links LLM scaling dynamics, capital accumulation, valuation growth, and bubble formation. This paper seeks to fill that gap by integrating empirical market evidence with a formal analytical model of compute-driven growth. We derive conditions under which AI-related asset prices decouple from realizable productivity gains, thereby satisfying the defining characteristics of an asset bubble.

II. RELATED WORK

A substantial body of literature examines the interaction between technological change, asset valuation, and macroeconomic dynamics. One prominent strand focuses on the formation of asset bubbles and the role of expectations in driving prices away from fundamentals. Kindleberger and Aliber characterize speculative episodes as recurring phenomena triggered by displacement events—often technological innovations—that alter expectations of future returns and initiate boom–bust cycles [3]. Minsky's financial instability hypothesis complements this view by emphasizing endogenous risk accumulation during prolonged expansions, in which rising optimism encourages leverage and fragility within the financial system [4].

Shiller extends these frameworks by highlighting the importance of narratives in shaping investor behaviour. In his work on narrative economics, Shiller argues that popular stories surrounding transformative technologies can spread contagiously and reinforce speculative dynamics even in the absence of commensurate growth in underlying cash flows [2], [34]. Empirically, Shiller's cyclically adjusted price–earnings (CAPE) ratio has been widely adopted as a long-horizon valuation benchmark, with values exceeding 30 historically associated with periods of speculative excess followed by major market corrections [1], [10]. Historical analyses support this interpretation: DeLong and Shleifer document how extrapolative expectations during the late 1920s preceded an 89% collapse in the Dow Jones Industrial Average, while the dot-com episode saw the S&P 500 CAPE ratio reach an all-time high of 44.19 before a sharp market reversal [9], [10].

A related literature examines the macroeconomic impact of general-purpose technologies (GPTs) and the timing of productivity gains. David's seminal analysis of electrification demonstrates that large productivity improvements often lag technological invention by decades, as firms require complementary organizational and institutional adjustments to fully exploit new technologies [14]. Gordon similarly argues that many modern innovations yield more limited aggregate productivity gains than earlier industrial breakthroughs, contributing to slower long-run growth [13]. More recent work identifies a persistent productivity paradox in the digital era: Brynjolfsson et al. show that while firm-level productivity gains from digital technologies can be substantial, aggregate productivity growth may remain muted due to adjustment costs, reallocation frictions, and measurement challenges [15]. OECD studies further emphasize that productivity gains from advanced technologies depend critically on complementary investments in skills, data governance, and organizational capital, leading to highly uneven diffusion across firms and sectors [16], [30].

Recent research on artificial intelligence bridges these literatures by examining both technical scaling and economic implications. In machine learning, empirical scaling laws show that model performance improves predictably with increases in compute, data, and parameter count, albeit with diminishing marginal returns [31]. Hoffmann et al. refine these results by demonstrating that compute-optimal training requires balancing model size and data rather than indiscriminate scaling [32]. While these findings are central to AI engineering, their economic interpretation remains ambiguous, as performance improvements on benchmarks do not directly translate into proportional gains in productivity, revenue, or consumer surplus.

Economic analyses of AI adopt a more cautious perspective. Acemoglu and Restrepo argue that the contribution of AI to aggregate growth is constrained by task structure, demand elasticity, and distributional effects, limiting the extent to which automation can sustain long-run productivity growth [17]. Empirical studies further show that AI adoption is uneven across firms, with large organizations facing greater integration, governance, and coordination frictions than smaller firms [18]–[21]. Industry data indicate that AI production is exceptionally capital intensive, requiring sustained investment in specialized hardware, energy infrastructure, and long-term cloud capacity contracts [22]–[26]. Capital-flow analyses reveal dense interdependencies between AI developers, cloud providers, and semiconductor manufacturers, suggesting that shocks to any segment of the AI supply chain may propagate across the broader technology and financial system [27].

A recent macroeconomic critique of AI optimism is provided by Storm, who argues that prevailing narratives of AI-driven growth overstate productivity gains while underestimating capital intensity, demand-side constraints, and distributional effects [33]. This perspective aligns with earlier theories of speculative dynamics, which emphasize the interaction between technological narratives, capital concentration, and financial markets [3], [34], [35]. However, while existing work offers important insights into bubbles, productivity diffusion, and AI scaling, these strands remain largely disconnected in formal analysis.

Table I summarizes key contributions from the related literature, highlighting their core findings, quantitative benchmarks, and limitations with respect to explaining valuation dynamics in the context of large-scale AI deployment.

III. MACROECONOMIC FRAMEWORK

In this section, we develop a stylized macroeconomic framework to analyse how large language model (LLM) scaling translates into economic output, capital accumulation, and asset valuation. The objective is not to model the full complexity of AI production, but to capture the key structural features that distinguish LLM-driven growth from prior software-based technologies: extreme capital intensity, diminishing returns to compute, and delayed diffusion into the real economy.

A. Production Strucutre

Consider an economy in discrete time t=0,1,2…populated by firms that employ labor and AI-enabled capital to produce output. Aggregate output Y_t is given by

$$Y_t = A_t L_t + \Phi g(C_t)$$

where A_t denotes baseline total factor productivity unrelated to AI, L_t is effective labor input, C_t represents the aggregate stock of AI-related compute deployed in production, $g()$ captures the contribution of AI to output, and Φ>0 scales the economic relevance of AI-generated output.

Consistent with empirical scaling laws observed in large language models, the contribution of AI to output is assumed to exhibit diminishing returns to compute:

$$g(C_t) = C_t^{\{\alpha\}}, 0 < \alpha < 1$$

TABLE I : Literature Comparison

Ref.	Literature Stream	Core Result	Quantitative / Empirical Insight	Limitation
[1], [10]	Asset valuation	Valuation extremes precede crashes	CAPE > 30 before 1929, 2000	No technology-specific mechanism
[3], [4]	Bubble theory	Speculation amplifies instability	Recurrent boom–bust cycles	Lacks sectoral structure
[9]	Historical bubbles	Extrapolative expectations dominate	1929 crash: −89% DJIA	Descriptive
[14]	GPT diffusion	Productivity gains delayed	Decades-long lag	No valuation link
[15]	Productivity paradox	Firm gains ≠ aggregate gains	Weak post-2000 productivity	No finance channel
[17]	AI and growth	AI growth constrained	Task and demand limits	No asset pricing
[31]	ML scaling laws	Performance scales with compute	Diminishing returns	No economic surplus
[32]	Compute-optimal scaling	Efficient scaling exists	Capital efficiency bounds	Ignores markets
[18]–[21]	AI adoption	Large-firm adoption plateaus	Declining adoption rates	Descriptive
[22]–[26]	AI capital intensity	High fixed costs	Rising capex and energy use	No dynamics
[33]	Macro critique	AI hype overstates growth	Demand-side constraints	No formal model

B. Compute Accumulation and Captial Intensity

The stock of AI-related compute evolves according to

$$C_{\{t+1\}} = (1 - \delta)C_t + I_t$$

Where $I_t > 0$, denotes gross investment in AI compute infrastructure and $\delta \in (0,1)$ is the depreciation rate of compute capital.

C. Profit and Value Capture

Let Π_t denote aggregate profits generated by AI-enabled production. Profits are assumed to be a constant fraction of AI-generated output:

$$\Pi_t = \theta\Phi\, C_t^{\{\alpha\}}$$

where $\theta \in (0,1)$ represents the share of AI output captured as profits by AI-related firms.

D. Asset Valuation

Let V_t denote the market valuation of AI-related firms. Under standard asset-pricing assumptions, valuation equals the expected discounted stream of future profits:

$$V_t = E_t \sum_{k=0}^{\infty} \beta^k \Pi_{(t+k)}$$

where $\beta \in (0,1)$ is the discount factor and expectations are taken with respect to information available at time t.

Substituting the profit expression yields

$$V_t = \theta\Phi E_t \sum_{k=0}^{\infty} \beta^k C_{(t+k)}^{\alpha}$$

E. Adoption Frictions

Effective utilization of compute capital is subject to adoption frictions. Let effective compute be defined as

$$C_t = \lambda_t C_t$$

where $\lambda_t \in (0,1]$ represents the degree of organizational and institutional adoption.

The adoption process evolves according to

$$\lambda_{(t+1)} = \lambda_t + \kappa(\lambda_t - \lambda_t), 0 < \kappa \ll 1$$

with $\lambda_t \leq 1$ denoting the long-run adoption ceiling.

F. Valuation Dynamics and Fundamental Bounds

The growth rate of market valuation is defined as

$$G_t^V = \frac{d}{dt} \ln V_t$$

and define fundamental growth as the growth rate of expected realizable profits,

$$G_t^F = \frac{d}{dt} ln\left(E_t \sum_{k=0}^{\infty} \beta^k C_{(t+k)}^{\alpha}\right)$$

Assume that expected compute capital grows at a finite rate g_C, such that

$$E_t\left[\frac{d}{dt} ln\, C_{(t+k)}\right] = g_C, \qquad k \geq 0$$

Given diminishing returns to compute, the growth rate of AI-generated output satisfies

$$\frac{d}{dt} ln\, C_{(t+k)}^{\alpha} = \alpha g_C, 0 < \alpha < 1$$

Effective utilization further constrains realizable surplus. Let the adoption rate satisfy

$$\frac{d}{dt} ln\, \lambda_t = g_\lambda, \quad g_\lambda \ll g_C$$

The growth rate of realizable profits is therefore bounded above by

$$G_t^F \leq \alpha g_C + g_\lambda$$

A valuation–fundamental disconnect arises whenever valuation growth exceeds this bound, i.e.,

$$G_t^V > G_t^F$$

This condition characterizes circumstances under which asset prices of AI-related firms grow faster than the maximum rate consistent with realizable productivity and profit generation.

IV. NUMERICAL SIMULATIONS AND SCENARIO ANALYSIS

This section presents a set of reduced-form simulations designed to illustrate the implications of the macroeconomic model under empirically plausible assumptions. Rather than solving the analytical framework in closed form, the simulations specify calibrated functional relationships governing large language model (LLM) scaling, capital allocation, costs, and systemic risk. These relationships are intended to capture stylized facts observed in AI deployment, investment behavior, and financial markets.

The reduced-form relationships are constructed to capture key stylized facts observed in recent AI deployment cycles, including diminishing marginal returns to compute, rapid capital deepening with delayed productivity effects, increasing concentration and financial interdependence within the AI supply chain, and the emergence of physical and balance-sheet constraints

A. Diminishing Returns to LLM Scaling

LLM performance is modeled as a concave function of compute scale C, reflecting diminishing marginal returns observed in empirical scaling studies:

$$Performance(C) = A[ln(C)]^{\alpha}, 0 < \alpha < 1$$

The logarithmic specification ensures that successive increases in compute yield progressively smaller performance improvements. The associated marginal product of compute is given by

$$MP(C) = (\partial Performance)/(\partial C) = \left(A\alpha[ln(C)]^{(\alpha-1)}\right)/C$$

This formulation implies that marginal gains decline at a rate $1/C$, such that exponential growth in compute produces sublinear improvements in model performance.

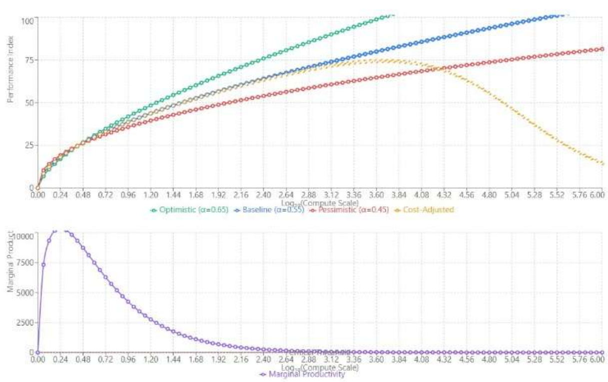

Figure 5: Diminishing returns to large language model scaling, showing sublinear performance gains and rapidly declining marginal productivity as compute expands.

B. Growth Accounting with AI Capital

Aggregate output follows a Cobb–Douglas specification:

$$Y(t) = A(t)K(t)^{\beta}L(t)^{(1-\beta)}$$

Total factor productivity evolves according to

$$A(t) = A_0e^{(g_A t)} + \delta ln(K_A I/K_0)$$

allowing AI capital to affect productivity through a weak logarithmic spillover rather than a discrete regime shift. Labor productivity is therefore given by

$$Y/L = A(t)(K/L)^{\beta}$$

implying that rapid capital deepening driven by AI investment need not translate into commensurate productivity gains.

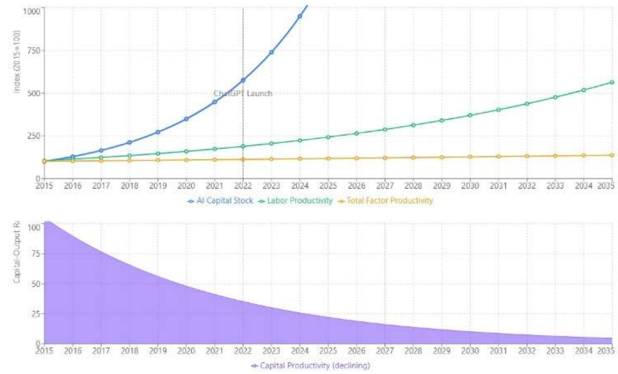

Figure 6: Rapid AI capital accumulation accompanied by weak growth in labour productivity and total factor productivity, indicating limited spillovers from compute-intensive investment.

C. Investment Crowding-Out Dynamics

Total investment is allocated across AI and non-AI sectors:

$$I_{total} = I_{AI} + I_M + I_{INF} + I_G$$

AI investment follows an exponential trajectory,

$$I_{AI}(t) = \alpha_0 e^{(\gamma_{AI} t)}$$

while investment in other sectors declines according to

$$I_j(t) = \beta_j e^{(-\varepsilon_j t)}$$

Normalized investment shares are computed as

$$Share_i(t) = \left(I_i(t)\right)/\left(I_{total}(t)\right) \times 100$$

The opportunity cost of reallocation is defined as

$$OC(t) = \sum_j w_j\left[I_j(2015) - I_j(t)\right]$$

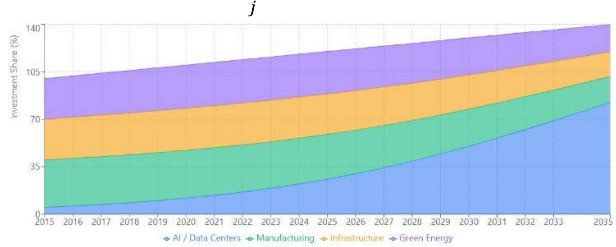

Figure 7: Rising allocation of aggregate investment toward AI infrastructure, crowding out manufacturing, infrastructure, and green energy sectors over time.

D. Circular Financial Flows and Systemic Risk

Financial interdependence increases over time according to

$$FI(t) = min(0.9, \rho_0 + \gamma t)$$

The probability of cascade failures is modeled as

$$P_{cascade}(t) = 1 - e^{(-\lambda L(t)t)}$$

A composite systemic risk index is constructed as

$$SRI(t) = 0.30FI + 0.25L + 0.20HHI + 0.25P_{cascade}$$

E. Revenue-Cost Dynamics and Profitability

Revenue evolves with saturation effects

$$R(t) = R_0 e^{(g_r t)}/(1 + \sigma t)(1 + \pi t)$$

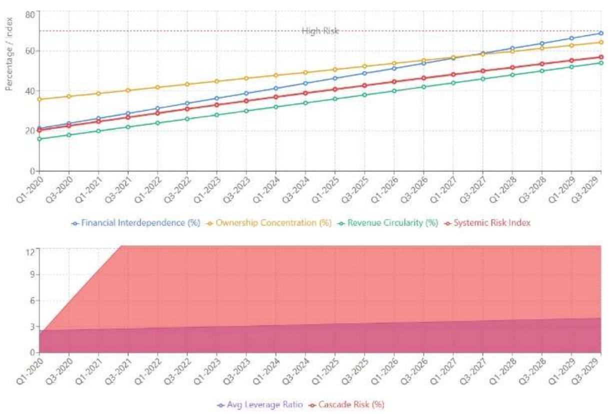

Figure 8: Increasing financial interdependence and concentration driving a composite systemic risk index toward elevated risk levels.

Total costs are given by

$$C_{total} = C_I + C_E + C_D$$

Break-even scaling is defined as

$$BES(t) = \left(C_total(t)\right)/\left(R(t)\right)$$

When BES>1, the business model becomes structurally unprofitable, regardless of scale.

F. Energy Constraints

Energy demand grows super-linearly with compute:

$$E(C) = \varepsilon C^{\theta}, \theta > 0.5$$

Grid capacity expands slowly:

$$G(C) = G_0 + kl\,n(C+1)$$

Utilization is given by

$$U(C) = \left(E(C)\right)/\left(G(C)\right) \times 100$$

When U>100%, AI scaling hits a hard physical constraint.

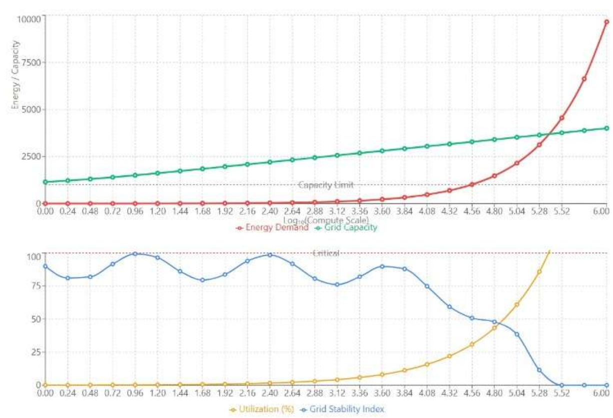

Figure 9: Super-linear growth in energy demand from AI scaling exceeding grid capacity and reducing system stability at high utilization rates.

G. Labour Market Effects

Wage dynamics by skill group satisfy

$$w_H(a) = w_0 + \beta_H a, w_M(a) = w_0 - \beta_M a,$$
$$w_L(a) = w_0 - \beta_L a$$

Employment and inequality evolve as

$$E_H(a) = 100 + 0.15a, E_M(a) = 100 - 0.12a$$
$$Gini(a) = 0.35 + 0.15a$$

AI increases inequality even when average productivity rises.

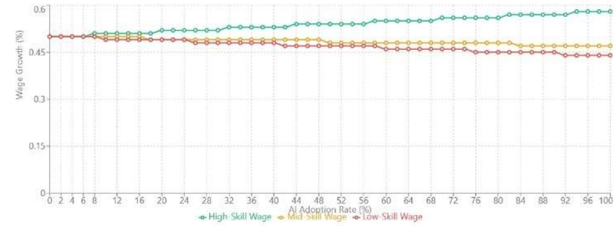

Figure 10: Divergent labor market outcomes from AI adoption, with rising high-skill wages and declining mid- and low-skill wage growth, increasing inequality.

H. Capital Obsolescence and Leverage

AI capital depreciates rapidly:

$$V_A I(t) = V_0 e^{(-0.35\ )}$$

Leverage and default risk evolve as

$$L(t) = \left(D(t)\right)/\left(V_A I(t)\right)$$

$$P_{default}(t) = 1 - e^{(-\mu t L(t))}$$

This formalizes balance-sheet fragility from rapid obsolescence.

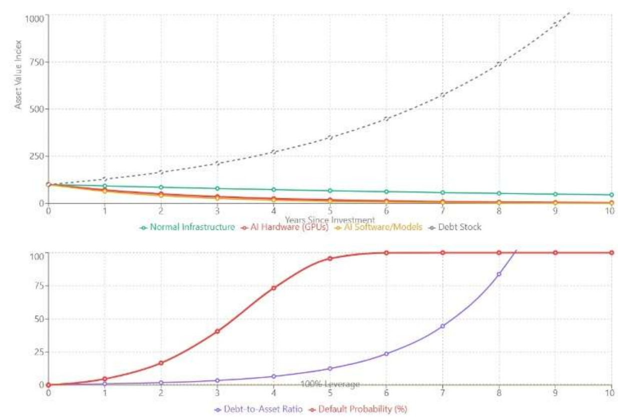

Figure 11: Accelerated obsolescence of AI capital leading to rising leverage ratios and sharply increasing default risk.

I. Bubble Indicator Index

A composite bubble indicator is defined as

$$BI(t) = 0.30Valuation + 0.25Leverage$$
$$+ 0.20Sentiment$$
$$+ 0.25FundamentalGap$$

Values exceeding the threshold indicate a bubble regime.

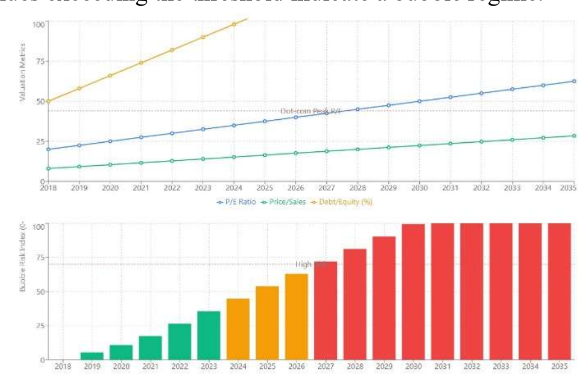

Figure 12: Expansion of valuation multiples and leverage pushing a composite AI bubble indicator beyond historical risk thresholds.

J. Competing Technology Paths

Generic LLM scaling follows

$$O_{LLM}(t) = O_0[l\,n(I_{LLM} + 1)]^{\alpha}$$

while targeted AI evolves as

$$O_T(t) = O_0 + \beta t + \gamma l\,n(I_T)$$

Return on investment is

$$ROI(t) = \big(O(t)\big)/\big(I(t)\big)$$

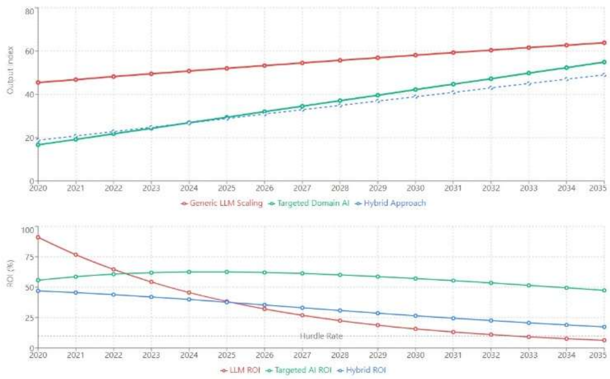

Figure 13: Comparison of AI development strategies showing higher long-run output stability and returns for targeted and hybrid approaches relative to generic LLM scaling.

V. RESULTS AND DISCUSSION

The numerical simulations reveal a coherent set of macroeconomic and financial dynamics that emerge from compute-intensive large language model (LLM) scaling. Taken together, the figures illustrate how diminishing returns to compute, extreme capital intensity, delayed productivity diffusion, and increasing financial interdependence interact to generate sustained valuation pressures and systemic risk.

At the technological level, the simulations show that performance improvements from LLM scaling follow a strongly concave trajectory. As illustrated in Figure 5, model performance continues to increase with compute, but marginal gains decline rapidly across all scenarios. This result establishes a fundamental limit to compute-led growth: even under optimistic assumptions, successive increases in capital expenditure deliver progressively smaller improvements in capability. As a consequence, expectations of unbounded productivity gains from continued scaling are not supported by the simulated performance dynamics.

This technological constraint translates directly into a macroeconomic disconnect between capital accumulation and productivity outcomes. Despite rapid growth in AI-related capital, labour productivity and total factor productivity rise only modestly and with substantial delay, as shown in Figure 6. The divergence reflects weak spillovers from compute-intensive investment into economy-wide production, consistent with historical productivity paradoxes associated with general-purpose technologies. The simulations indicate that capital deepening alone is insufficient to generate sustained productivity growth in the absence of complementary organizational and institutional adjustments.

The reallocation of capital toward AI infrastructure further amplifies this disconnect. Figure 7 shows a persistent shift of aggregate investment toward AI data centers at the expense of manufacturing, physical infrastructure, and green energy. While total investment rises, the composition of capital formation becomes increasingly concentrated in assets characterized by high fixed costs, rapid obsolescence, and limited cross-sector spillovers. This crowding-out effect introduces a macroeconomic opportunity cost, as capital is diverted away from sectors with historically higher multipliers and broader productivity externalities.

At the financial level, increasing concentration and interdependence within the AI ecosystem generate rising systemic fragility. As shown in Figure 8, cross-ownership, revenue circularity, and market concentration jointly push a composite systemic risk index toward elevated levels. The simulations suggest that valuation shocks or demand shortfalls affecting one segment of the AI supply chain may propagate rapidly across semiconductor manufacturers, cloud providers, and model developers. This tightly coupled structure magnifies downside risks even in the absence of an external shock.

Physical constraints reinforce these dynamics. Energy demand associated with AI scaling grows super-linearly with compute, while grid capacity expands only gradually. As depicted in Figure 9, utilization approaches critical thresholds at higher levels of compute deployment, implying that further scaling may require disproportionate investment in energy infrastructure or result in sharply rising operating costs. These constraints operate independently of financial considerations and further limit the sustainability of compute-driven expansion.

Distributional effects in labor markets compound the macroeconomic challenges. The simulations indicate that AI adoption disproportionately benefits high-skill labor while reducing wage growth for mid- and low-skill workers, leading to rising inequality even as aggregate output increases. This divergence has implications for aggregate demand and political economy, potentially constraining the realization of productivity gains at the macro level.

Financial fragility is further exacerbated by rapid capital obsolescence. As illustrated in Figure 11, AI-related assets depreciate substantially faster than traditional infrastructure, while debt obligations adjust more slowly. The resulting increase in leverage and default probabilities introduces a novel source of balance-sheet risk, particularly in environments characterized by aggressive expansion and optimistic valuation assumptions.

When these mechanisms are considered jointly, the valuation dynamics become clear. Rising capital expenditure, weak productivity spillovers, increasing leverage, and growing systemic risk coexist with rapidly expanding asset valuations. The composite bubble indicator shown in Figure 12 summarizes this interaction, crossing thresholds historically associated with valuation–fundamental divergence. Importantly, the simulations demonstrate that such dynamics can arise endogenously, without assuming irrational behaviour, purely from the structural features of AI-driven growth.

Finally, the comparison of alternative AI development paths highlights the role of capital efficiency. Targeted and hybrid AI strategies generate more stable output growth and superior long-run returns than generic large-scale LLM scaling, suggesting that breadth of deployment and integration with complementary investments matter more than raw compute intensity. This result reinforces the central finding of the paper: scale alone is not a sufficient driver of sustainable economic growth.

Overall, the results indicate that the current AI investment cycle exhibits several structural characteristics historically associated with speculative episodes: diminishing marginal returns, capital concentration, delayed productivity realization, and increasing financial fragility. While AI technologies may ultimately deliver significant long-run benefits, the simulations suggest that prevailing valuation dynamics are vulnerable to correction if expectations continue to outpace realizable economic gains.

VI. CONCLUSION

This paper examined the macroeconomic and financial implications of large language model (LLM)–driven growth through a unified analytical and simulation-based framework. By integrating diminishing returns to compute, capital intensity, delayed productivity diffusion, and financial interdependence, the analysis provides a structured explanation for the divergence between rapidly rising AI-related valuations and more modest realizable economic gains.

The results indicate that performance improvements from LLM scaling follow a concave trajectory, with marginal gains declining rapidly as compute expands. While AI-related capital accumulation accelerates following the deployment of frontier models, productivity growth responds slowly and unevenly, consistent with historical productivity paradoxes observed for earlier general-purpose technologies. At the same time, capital reallocation toward AI infrastructure crowds out investment in sectors with broader spillovers, potentially weakening long-run growth capacity despite rising aggregate investment.

On the financial side, increasing concentration, cross-ownership, and revenue circularity within the AI ecosystem amplify systemic fragility. Rapid technological obsolescence further erodes balance sheets by accelerating asset depreciation relative to debt adjustment, raising leverage and default risk even in the absence of adverse shocks. Physical constraints, particularly energy infrastructure limitations, introduce additional non-financial bounds on sustained scaling. Together, these mechanisms generate valuation dynamics that can persistently outpace realizable fundamentals, producing conditions historically associated with speculative episodes.

Importantly, the analysis shows that such dynamics do not require irrational expectations or exogenous disturbances. Instead, valuation–fundamental divergence arises endogenously from the structural characteristics of compute-intensive AI production. The composite bubble indicator constructed in the simulations reflects this interaction, summarizing how technological limits, capital allocation, and financial structure jointly elevate valuation risk.

The comparison of alternative AI development strategies highlights an important qualification to the prevailing scale-centric narrative. Targeted and hybrid AI approaches, which emphasize domain-specific deployment and complementary investments, deliver more stable output growth and higher long-run returns than generic large-scale LLM scaling. This suggests that capital efficiency and integration matter more for sustainable growth than sheer compute intensity.

Several limitations warrant acknowledgment. The simulation framework is reduced-form and calibrated to stylized facts rather than micro founded behavioural equations. Data limitations constrain the empirical validation of certain mechanisms, particularly those related to financial interdependence and energy infrastructure. Future research could extend the framework by incorporating endogenous demand responses, heterogeneous firm behaviour, and explicit policy regimes, as well as by validating the model against firm-level and sectoral data as AI adoption matures.

Overall, the findings caution against interpreting rapid AI-driven valuation growth as a direct indicator of imminent economy-wide productivity gains. While AI technologies are likely to deliver substantial long-run benefits, the path to those gains is shaped by capital intensity, institutional adaptation, and physical constraints. Recognizing these dynamics is essential for investors, policymakers, and firms seeking to navigate the current AI investment cycle without repeating the excesses of past technology-led booms.

REFERENCES

[1] R. Shiller, *Irrational Exuberance*, 3rd ed. Princeton, NJ, USA: Princeton Univ. Press, 2015.

[2] R. J. Shiller, "Narrative economics," *Amer. Econ. Rev.*, vol. 107, no. 4, pp. 967–1004, 2017, doi: 10.1257/aer.107.4.967.

[3] C. P. Kindleberger and R. Z. Aliber, *Manias, Panics, and Crashes*, 7th ed. London, U.K.: Palgrave Macmillan, 2011, doi: 10.1057/9780230362595.

[4] H. P. Minsky, *Stabilizing an Unstable Economy*. New Haven, CT, USA: Yale Univ. Press, 1986.

[5] S&P Dow Jones Indices, "Market concentration and index dominance," 2024.

[6] Macrotrends, "Historical stock valuation data," 2024.

[7] Motley Fool, "The Magnificent Seven and S&P 500 concentration," 2024.

[8] Financial Times, "U.S. stock market concentration reaches historic highs," 2024.

[9] J. B. DeLong and A. Shleifer, "The stock market bubble of 1929," *J. Econ. Hist.*, vol. 51, no. 3, pp. 675–700, 1991, doi: 10.1017/S002205070003915X.

[10] R. J. Shiller, "Measuring bubble expectations," *Quart. J. Econ.*, vol. 129, no. 4, pp. 1531–1575, 2014, doi: 10.1093/qje/qju023.

[11] A. Tobin, "A general equilibrium approach to monetary theory," *J. Money Credit Banking*, vol. 1, no. 1, pp. 15–29, 1969.

[12] L. Summers, "Secular stagnation," *Amer. Econ. Rev.*, vol. 106, no. 5, pp. 503–507, 2016, doi: 10.1257/aer.p20161023.

[13] R. Gordon, *The Rise and Fall of American Growth*. Princeton, NJ, USA: Princeton Univ. Press, 2016.

[14] P. A. David, "The dynamo and the computer," *Amer. Econ. Rev.*, vol. 80, no. 2, pp. 355–361, 1990.

[15] E. Brynjolfsson, D. Rock, and C. Syverson, "Artificial intelligence and the modern productivity paradox," *J. Econ. Persp.*, vol. 35, no. 4, pp. 1–24, 2021, doi: 10.1257/jep.35.4.1.

[16] OECD, *AI, Productivity and the Firm*, Paris, France, 2023.
[17] Acemoglu and Restrepo, "AI and jobs," *J. Econ. Persp.*, vol. 33, no. 2, pp. 3–30, 2020, doi: 10.1257/jep.33.2.3.
[18] Financial Times, "Why AI adoption is stalling in large firms," 2024.
[19] Wall Street Journal, "When AI hype meets AI reality," 2024.
[20] Apollo Academy, "AI adoption rate trending down for large companies," 2024.
[21] McKinsey Global Institute, *The State of AI*, 2023.
[22] Bloomberg, "AI capital expenditures surge," 2024.
[23] NVIDIA Corp., *Form 10-K*, 2024.
[24] Microsoft Corp., *Form 10-K*, 2024.
[25] Oracle Corp., *Form 10-K*, 2024.
[26] International Energy Agency, "Energy implications of data centers," 2023.
[27] Bloomberg, "Mapping AI capital flows," 2024.
[28] D. Autor, "Why are there still so many jobs?" *J. Econ. Persp.*, vol. 29, no. 3, pp. 3–30, 2015, doi: 10.1257/jep.29.3.3.
[29] Acemoglu et al., "AI and economic growth," NBER Working Paper, 2022.
[30] OECD, *Employment Outlook*, 2023.
[31] J. Kaplan et al., "Scaling laws for neural language models," arXiv:2001.08361, 2020.
[32] J. Hoffmann et al., "Training compute-optimal large language models," arXiv:2203.15556, 2022.
[33] S. Storm, "The myth of AI-driven productivity growth," INET Working Paper No. 244, 2024, doi: 10.36687/inetwp244.
[34] R. Shiller, *Narrative Economics*. Princeton, NJ, USA: Princeton Univ. Press, 2019.
[35] G. Akerlof and R. Shiller, *Animal Spirits*. Princeton, NJ, USA: Princeton Univ. Press, 2009, doi: 10.1515/9781400834181.